MSC MARRIAGE AND FAMILY THERAPY

LECTURE LIBRARY · VOLUME 21 OF 28

# MFT 721

# Research Methodology

14 teaching weeks · 2 practicals · 2 figures

SEMESTER THREE

1.71 units · Prerequisite: MFT 705

> “This is the advanced methods course that turns a proposal into a thesis.”

One of 28 self-contained course books in the MSc Marriage and Family Therapy lecture library: a full taught semester per volume, with cases, practicals, assessment briefs, glossaries and model answers.



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# MFT 721 — Research Methodology

**MSc Marriage and Family Therapy** · Semester Three · 1.71 Units · 14 teaching weeks · Prerequisite: MFT 705

This volume is one of 28 course books that together cover the whole MSc Marriage and Family Therapy taught programme. Each book is written to carry a 14–16 week semester on its own: every week pairs a core idea with the practice that follows from it, a worked case, and a reflective task, and each book closes with an integration chapter, a glossary and a self-assessment paper with model answers.

The books are designed to be read straight through before the semester begins, revisited week by week, and used again in revision. Cross-references point to the other volumes where a topic is treated in more depth.


#### How to cite this volume

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#### A note on the case material

All families, couples and individuals in this book are fictional composites. They are written to teach pattern recognition, not to describe any real person. Nothing here replaces supervised clinical practice, professional registration, legal advice or medical care; legal content is orientation only and must be checked against the law of the relevant jurisdiction.

# The lecture library

Twenty-eight self-contained course books carry the taught programme: eight in Semester One, seven in Semester Two, seven in Semester Three including the thesis, and six electives. The volume you are holding is marked in the margin.

| Code | Course | Semester | Units |
|---|---|---|---|
| MFT 701 | Fundamentals of Marriage and Family Therapy | One | 1.71 |
| MFT 702 | Theories of Family Therapy | One | 1.71 |
| MFT 703 | Human Development Across the Lifespan | One | 1.71 |
| MFT 704 | Ethical and Legal Issues in Therapy | One | 1.71 |
| MFT 705 | Research Methods in Marriage and Family Therapy | One | 1.71 |
| MFT 706 | Clinical Practice and Techniques | One | 1.71 |
| MFT 707 | Assessment and Diagnosis in Family Therapy | One | 1.71 |
| MFT 708 | Cultural Competence in Therapy | One | 1.71 |
| MFT 709 | Group Therapy Techniques | Two | 1.71 |
| MFT 710 | Couples Therapy | Two | 1.71 |
| MFT 711 | Child and Adolescent Therapy | Two | 1.71 |
| MFT 712 | Crisis Intervention Strategies | Two | 1.71 |
| MFT 713 | Substance Abuse and Family Dynamics | Two | 1.71 |
| MFT 714 | Trauma-Informed Care | Two | 1.71 |
| MFT 715 | Family Systems Theory | Two | 1.71 |
| MFT 716 | Communication Skills in Therapy | Three | 1.71 |
| MFT 717 | Psychopathology in Families | Three | 1.71 |
| MFT 718 | Intimacy and Relationship Dynamics | Three | 1.71 |
| MFT 719 | Family Life Cycle and Transitions | Three | 1.71 |
| MFT 720 | Supervision and Consultation in Therapy | Three | 1.71 |
| **MFT 721** | **Research Methodology** | Three | 1.71 |
| MFT 722 | Professional Development and Self-Care | Electives | 1.71 |
| MFT 723 | Special Populations in Therapy | Electives | 1.71 |
| MFT 724 | Family Law and Advocacy | Electives | 1.71 |
| MFT 725 | Integrative Approaches in Therapy | Electives | 1.71 |
| MFT 726 | Neurobiology of Relationships | Electives | 1.71 |
| MFT 727 | Play Therapy Techniques | Electives | 1.71 |
| MFT 770 | Master Thesis | Three | 1.71 |

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**Course leader's summary.** This is the advanced methods course that turns a proposal into a thesis. You will work with real data, choose and justify analytic strategies, defend your design against examiners, and learn to write findings that practitioners will actually read.

## How to use this book

Every week's task contributes directly to your thesis. Bring your own dataset, or use the teaching dataset supplied with the course. Do not read statistics passively; run them, break them, and interpret them badly at least once.

## 1. Course description

The course covers advanced design (longitudinal, dyadic, single-case, participatory), analysis (regression, multilevel modelling, dyadic models, thematic and framework analysis, mixed-methods integration), measurement and psychometric evaluation, systematic reviewing, data management and reproducibility, writing for publication, and defence of a study.

## 2. Learning outcomes

| # | Outcome | Assessed |
|---|---|---|
| LO1 | Select and justify a design for a complex relational question | Thesis proposal |
| LO2 | Conduct and interpret appropriate analyses for relational data | Analysis portfolio |
| LO3 | Evaluate the psychometric quality of a measure in a new population | Psychometric report |
| LO4 | Integrate quantitative and qualitative findings coherently | Mixed-methods report |
| LO5 | Defend design decisions against alternative explanations | Viva |
| LO6 | Write a publishable findings chapter | Thesis chapter |
| LO7 | Manage data ethically, securely and reproducibly | Data plan and audit |

## 3. Assessment

| Component | Weight | Due |
|---|---|---|
| Analysis portfolio (4 exercises) | 30% | Weeks 5, 8, 11, 13 |
| Psychometric evaluation report (1,500 words) | 20% | Week 9 |
| Mixed-methods integration report (2,000 words) | 20% | Week 12 |
| Thesis methodology chapter + viva | 30% | Week 15 |

## 4. The fourteen-week plan

| Week | Topic | Task |
|---|---|---|
| 1 | From proposal to protocol; pre-registration and its limits | Protocol draft |
| 2 | Design for relational questions | Design decision log |
| 3 | Sampling, power and feasibility in clinical settings | Power calculation |
| 4 | Data preparation and the analysis plan | Data cleaning log |
| 5 | **Analysis 1:** regression and its assumptions | Run and interpret |
| 6 | Multilevel models for nested family data | Model comparison |
| 7 | Dyadic models: APIM in practice | APIM exercise |
| 8 | **Analysis 2:** thematic and framework analysis | Coding audit |
| 9 | Psychometrics: reliability, factor structure, invariance | Report |
| 10 | Systematic review and meta-analysis basics | Search and appraise |
| 11 | **Analysis 3:** single-case and small-N designs | SCED exercise |
| 12 | Mixed methods and integration | Joint display |
| 13 | **Analysis 4:** robustness, missing data, common pitfalls | Sensitivity analysis |
| 14 | Writing, publishing and defending | Chapter and viva prep |

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## Week 1 — Protocol and pre-registration

**The protocol** is the study's constitution: question, design, sample, measures, analysis plan, ethics, and timeline. Write it before you collect data; revise it with dated amendments if you must change course.

**Pre-registration** (e.g. on OSF) commits you publicly to a plan, which reduces the risk of unconscious analytical flexibility ("p-hacking"). It is standard in trials and increasingly expected elsewhere.

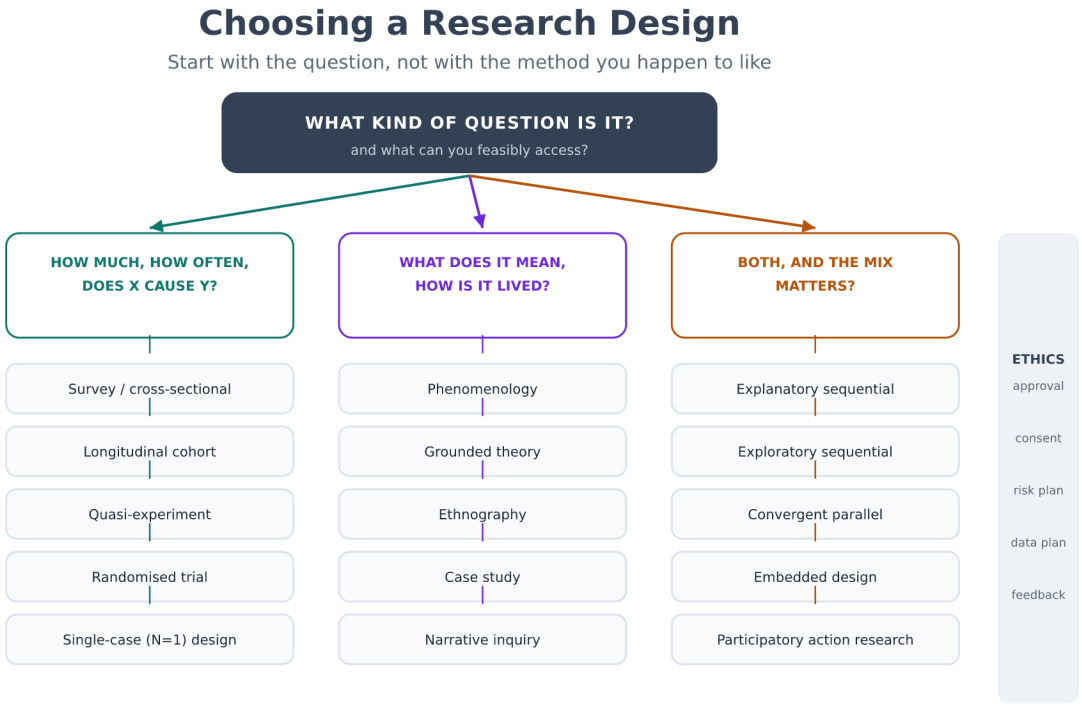
**Honest limits.** Exploratory qualitative work and community-based participatory research are often incompatible with rigid pre-registration. Declare your approach and be transparent about decisions made during the study.

**A live analysis log** — one page per week, dated — is the single most useful document in a thesis. It records what you did, what you decided, and why.

> **TASK**
> Draft a two-page protocol for your thesis. Include one paragraph on what you will do if recruitment is half of what you expect.

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# Week 2 — Design for relational questions




**NOTE**

In MFT the unit of analysis is the relationship, not only the individual — which creates real statistical problems: partners are non-independent. Plan for it (dyadic data analysis, actor–partner models, or a design that samples one person per household).

Figure 2.1 — Choosing a design from the question

### Four designs especially suited to family questions.

| Design | Question | Strength | Weakness |
|---|---|---|---|
| Longitudinal panel | Does X predict change in Y? | Temporal ordering | Attrition, cost |
| Dyadic (APIM) | How do partners affect each other? | Separates actor and partner effects | Requires both partners |
| Single-case experimental (SCED) | Does this intervention work for this family? | Clinical realism, causal inference within case | Generalisation |
| Participatory action research | What would improve this community's experience? | Validity, ethics, impact | Time, power negotiation |

**The design decision log.** For each choice, record: the alternative considered, the reason for rejection, and the threat it leaves unresolved. Examiners reward this; it shows you know what your design cannot do.

## Week 3 — Sampling, power and feasibility

**Power.** Do the calculation before you collect (GPower, R's `pwr`, or simulation). For dyadic and multilevel designs, power depends on the number of clusters* (couples, families), not the number of individuals.

**Practical realities in clinical settings.**

- Referral rates: how many families does the clinic actually see per week?
- Eligibility: what proportion will meet your criteria?
- Consent: what proportion will agree? (Often 30–60% in clinical samples.)
- Attrition: assume 20–30% at follow-up.
- Seasonality: school terms, holidays, religious calendars.

**The feasibility sentence.** "To recruit 60 couples in six months, I need 4 eligible referrals per week with a 60% consent rate; the clinic receives 9 per week; two colleagues have agreed to refer."

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## Week 4 — Data preparation

**The unglamorous work that determines everything.**

1. **Codebook** — every variable, its label, its values, its missingness convention.
2. **Raw data frozen** — never edit the original file after data entry.
3. **Cleaning log** — every decision (a value of 99 recoded to missing; one duplicate removed).
4. **Missing data** — describe the pattern; distinguish missing at random from missing not at random; consider multiple imputation rather than listwise deletion.
5. **Reverse scoring** — check; a mis-scored subscale invalidates the results.
6. **Outliers** — inspect, do not automatically delete.

> **TASK**
> Produce a codebook and cleaning log for your dataset (or the teaching dataset).

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## Week 5 — Regression

**What regression does.** Estimates the association between predictors and an outcome, adjusting for other predictors, and expresses it in the outcome's units.

**Assumptions to check.** Linearity, independence of errors, homoscedasticity, normality of residuals (for inference), and no severe multicollinearity. Report the checks; do not just assert them.

**Reading output.** Coefficient (direction, size, units), standard error, confidence interval, p-value, $R^2$, and model fit. Report effect sizes with confidence intervals, not only p-values.

**Clinical translation.** "Each additional point on the criticism scale was associated with a 0.3-point lower satisfaction score (95% CI −0.5 to −0.1), holding depression constant." That sentence is worth more than a table of asterisks.

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## Week 6 — Multilevel models

**The problem.** Individuals nested in couples, couples nested in therapists, repeated measures nested in people. Standard regression assumes independence and will produce confidence intervals that are too narrow.

**The multilevel solution.** Model the nesting explicitly:

- **Fixed effects** — the average relationship across the sample.
- **Random effects** — how intercepts (and slopes) vary across clusters.
- **ICC** — the proportion of variance attributable to the cluster. A high ICC means the clustering matters greatly and cannot be ignored.

**Model-building sequence.** Null model (ICC) → add level-1 predictors → add level-2 predictors → test random slopes → compare with likelihood ratio tests or information criteria (AIC/BIC).

**Report.** Always report the ICC, the number of clusters, and the estimation method.

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## Week 7 — Dyadic models

**APIM in four numbers.**

- **Actor effect (a):** my predictor on my outcome.
- **Partner effect (p):** my predictor on my partner's outcome.
- **Distinguishability:** whether partners are distinguishable (e.g. by gender or role) and whether effects can be constrained equal.
- **K:** the correlation between partners' residuals, which models the non-independence.

**Interpretation example.** In a study of 80 couples, criticism predicted lower satisfaction for the person giving it (actor effect) more strongly than for the person receiving it (partner effect). Clinical implication: the critic is also being harmed, which is a more useful reframe than "stop criticising your partner".

**Software.** R (`lme4`, `nlme`, `ape`), Mplus, SPSS MIXED.

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## Week 8 — Qualitative analysis

**Two workable approaches.**

| Approach | Logic | Output |
|---|---|---|
| Reflexive thematic analysis (Braun & Clarke) | Themes constructed from coding, with the analyst's role explicit | Rich thematic account |

| Approach | Logic | Output |
|---|---|---|
| Framework analysis (Ritchie & Spencer) | Matrix of cases × themes, often for applied policy questions | Structured matrix, comparable across cases |

**Rigour in practice.**

- Keep a **coding audit trail**: initial codes, revisions, discarded codes and why.
- Use **analytic memos** after each transcript.
- **Second-coder check** on at least 20% of data with a documented discussion of divergence.
- **Member reflection** where appropriate (not to validate themes, but to deepen them).
- **Reflexive journal** on your own reactions and assumptions.

**The common error** is presenting themes as "findings that emerged" from data, as though the analyst were not present. Themes are constructed; say so.

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## Week 9 — Psychometrics




Figure 9.1 — Validity, reliability and their limits

**A psychometric evaluation of a measure in your population.**

1. **Internal consistency** — Cronbach's α (and ω, which is more appropriate for congeneric items).

2. **Factor structure** — exploratory or confirmatory factor analysis; report fit indices (CFI, TLI, RMSEA, SRMR).
3. **Construct validity** — correlations with related and unrelated measures.
4. **Measurement invariance** — configural, metric, scalar, across groups (gender, language, age).
5. **Responsiveness** — change scores in a treated sample.

**A local validation study is a perfectly good thesis**, and often more useful to the field than another outcome study using an unvalidated instrument.

> **REPORT (20%)**
> A 1,500-word psychometric evaluation of one instrument, with recommendations.

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## Week 10 — Systematic review basics

**Steps.** Register the protocol (PROSPERO) → define the question (PICO) → search systematically (at least three databases; record the strategy) → screen (two reviewers; report agreement) → extract data → assess risk of bias (e.g. Cochrane RoB 2, or CASP for qualitative) → synthesise (narrative or meta-analytic) → assess certainty (GRADE).

**Meta-analysis, briefly.** Pool effect sizes with a random-effects model unless there is a justification for fixed effects; report heterogeneity ($I^2$, $\tau^2$); explore it with subgroup analysis or meta-regression if there are enough studies; assess publication bias with a funnel plot and Egger's test, understanding their limitations.

**What examiners look for.** A search strategy that could be reproduced; a bias assessment that is not perfunctory; and a conclusion proportionate to the evidence.

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## Week 11 — Single-case and small-N designs

**Why they matter in family therapy.** Clinical samples are small, families are heterogeneous, and randomised trials are expensive. Single-case designs give causal inference within a case.

**Core designs.**

| Design | Structure | Inference |
|---|---|---|
| A–B | Baseline, then intervention | Weak (confounded by time) |
| A–B–A | Baseline, intervention, withdrawal | Stronger if reversal occurs |
| Multiple baseline across behaviours | Staggered introduction | Strong; no withdrawal needed |
| Multiple baseline across participants | Staggered across families | Strong; realistic in clinics |

**Analysis.** Visual analysis (level, trend, variability) plus a quantitative index (e.g. Tau-U) and, ideally, systematic replication across cases.

**Reporting standard.** Follow the SCED reporting guidelines (e.g. SCRIBE), including baseline stability and treatment fidelity.

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## Week 12 — Mixed methods integration

**The integration question.** What does the combination tell you that neither strand could alone?

**Joint display format.**

| Quantitative finding | Qualitative finding | Convergence / divergence | Interpretation |
|---|---|---|---|
| Attendance predicted outcome (β = .32) | Families described "being known" by the therapist as decisive | Divergence in mechanism | Attendance may be a proxy for alliance |
| Drop-out 34% | Families cited transport and work shifts | Convergence | Structural barriers, not motivation |

**Avoid the two common failures.** Reporting two parallel studies with no integration; and using qualitative data merely to illustrate quantitative findings.

> **REPORT (20%)**
> A 2,000-word integration report with at least three joint display rows.

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## Week 13 — Robustness and pitfalls

**Pitfalls you will meet.**

| Pitfall | What it looks like | Fix |
|---|---|---|
| p-hacking | Many tests, reporting the significant ones | Pre-specify; adjust; report all |
| HARKing | Hypothesis stated after results known | Register the hypothesis |
| Overfitting | Model too complex for the sample | Simpler model; cross-validation |
| Ignoring clustering | Treating partners as independent | Multilevel or dyadic models |
| Selective reporting | Reporting one measure, ignoring another | Report all pre-specified outcomes |
| Causal language | "Caused" from cross-sectional data | Use associational language |

**Sensitivity analysis.** Run your main analysis again with a different reasonable decision (e.g. excluding an outlier, using a different missing-data approach) and report whether conclusions change. If they do, say so.

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## Week 14 — Writing, publishing and defending

**The findings chapter.** Present, do not interpret (that is the discussion's job). Structure by question. Report descriptive statistics before inferential. Use tables for numbers and prose for meaning.

**The discussion.** What you found → how it compares with the literature → mechanisms → limitations (specific, not generic) → implications for practice and policy → future research.

**Defending.** Prepare for four questions: (1) Why this design? (2) What are the threats to validity and how did you address them? (3) Why these analyses? (4) What can and cannot be concluded? Rehearse short answers.

**Publishing.** Choose a journal that publishes practitioner research; follow the author guidelines exactly; expect a rejection and a revision; respond to reviewers point by point.

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## Glossary

**APIM** — Actor–Partner Interdependence Model. **Cluster** — a grouping unit (couple, family, therapist) whose members are not independent. **GRADE** — a system for rating the certainty of evidence. **ICC** — intraclass correlation; the share of variance due to clustering. **Measurement invariance** — equivalence of a measure's functioning across groups. **Pre-registration** — public commitment to a study plan before data collection. **SCED** — single-case experimental design. **Sensitivity analysis** — re-analysis under different assumptions to test robustness.

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## Self-assessment

### Multiple choice

**1.** Treating partners as independent observations primarily affects: (a) external validity (b) standard errors and Type I error (c) reliability (d) sampling

**2.** The ICC indicates: (a) internal consistency (b) the proportion of variance due to clustering (c) effect size (d) power

**3.** A multiple baseline design is strong because: (a) it requires withdrawal (b) staggered introduction controls for time (c) it uses large samples (d) it is qualitative

**4.** Joint displays are used to: (a) present descriptive statistics (b) integrate quantitative and qualitative findings (c) assess bias (d) calculate power

**5.** Pre-registration mainly reduces: (a) attrition (b) analytical flexibility and p-hacking (c) cost (d) sample size

**6.** Measurement invariance concerns: (a) whether a measure behaves equivalently across groups (b) internal consistency only (c) sample size (d) randomisation

**7.** A sensitivity analysis tests: (a) whether conclusions hold under different assumptions (b) the sample's normality (c) the coding frame (d) the ethics

**8.** HARKing means: (a) hypothesising after results are known (b) random sampling (c) qualitative coding (d) data entry

**9.** In framework analysis the output is typically: (a) a case × theme matrix (b) a regression table (c) a factor loading matrix (d) a funnel plot

**10.** The most useful document in a thesis is: (a) the title page (b) a dated analysis log (c) the abstract (d) the reference list

### Answer key

1 b · 2 b · 3 b · 4 b · 5 b · 6 a · 7 a · 8 a · 9 a · 10 b

### Short answer

1. Explain why clustering matters for standard errors.
2. Distinguish actor and partner effects with an example.
3. List five elements of a systematic review protocol.
4. What is the difference between a sensitivity analysis and a subgroup analysis?
5. Name four threats to validity in a pre-post clinic study.

### Essay (choose one, 2,000 words)

1. Defend (or attack) the claim that randomised trials are the gold standard for family therapy research.
2. Present a design for a dyadic study of criticism and satisfaction, justifying every choice.
3. Critically appraise a systematic review of family interventions, focusing on method and certainty.

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## Further reading


- Kenny, D. A., Kashy, D. A., & Cook, W. L. (2006). Dyadic Data Analysis. Guilford.
- Hox, J., Moerbeek, M., & van de Schoot, R. (2017). Multilevel Analysis (3rd ed.). Routledge.
- Braun, V., & Clarke, V. (2022). Thematic Analysis: A Practical Guide. Sage.
- Higgins, J. P. T. et al. (Eds.). (2023). Cochrane Handbook for Systematic Reviews of Interventions.
- Kazdin, A. E. (2021). Single-Case Research Designs (3rd ed.). OUP.


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Next: MFT 770 — Master Thesis